Deep Learning-Based Image Classification of Cat Skin Diseases

A Comparative Study of ResNet50, MobileNetV2, and EfficientNetB0

Chesney Kayzle A. Bumakil & Kathrin Grace S. Omac

School of Information Technology, Mapúa University

SCHOOL OF INFORMATION TECHNOLOGY

INTRODUCTION

Skin problems are a common reason cats see a veterinarian, and many feline skin diseases look alike: hair loss, redness, scaling and crusting appear in several of them. Some, like ringworm and scabies, can also spread to people and other animals. Earlier studies showed that convolutional neural networks (CNNs) can classify photos of cat skin disease, but they tested only lightweight models, used single data splits, and did not report accuracy and computing cost together. This study compares one high-capacity CNN (ResNet50) with two lightweight CNNs (MobileNetV2 and EfficientNetB0), trained and tested under identical conditions on the same photos of three skin diseases and healthy skin.

OBJECTIVES

1. Evaluate each model's **classification performance** (accuracy, macro-F1, MCC, per-class and disease-vs-healthy screening) and **computational characteristics** (parameters, FLOPs, model size, inference time).
2. **Compare** the three models and test whether their differences in accuracy are statistically significant.
3. Identify the **strengths and weaknesses** of each model when performance and computing cost are considered together.

METHODOLOGY

Dataset: 360 photos, 4 balanced classes

Flea allergy dermatitis	Ringworm (dermatophytosis)	Scabies (notoedric mange)	Healthy skin
90 images	90 images	90 images	90 images

Procedure (identical for all three models)

1 · Collect
360 public photos from Kaggle and Roboflow, 90 per class

2 · Clean & validate
Label review, near-duplicate check (pHash), veterinarian validation

3 · Preprocess
Letterbox to square, resize to 224×224; augment training images only

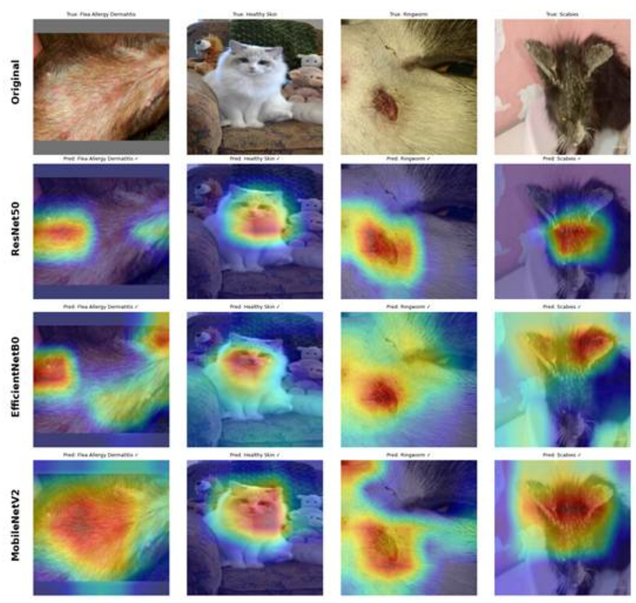
4 · Split
Stratified 5-fold cross-validation: 244 train, 44 validation, 72 test per fold

5 · Train
ImageNet weights; train head, then fine-tune last 30% of layers

6 · Evaluate
Performance, McNemar test with Holm correction, computing cost

A licensed veterinarian reviewed the dataset and judged it **valid with corrections**, rating healthy skin and scabies as the most reliable folders.

WHERE THE MODELS LOOKED



Grad-CAM heatmaps, fold 1. Rows: original, ResNet50, EfficientNetB0, MobileNetV2.

Heatmaps show which parts of a photo drove each prediction (red = strongest).

Ringworm: all three models focused on the lesion.

Scabies: EfficientNetB0 focused on the ears, where feline scabies typically appears.

Healthy skin: models looked at the face and chest, so framing may play a part.

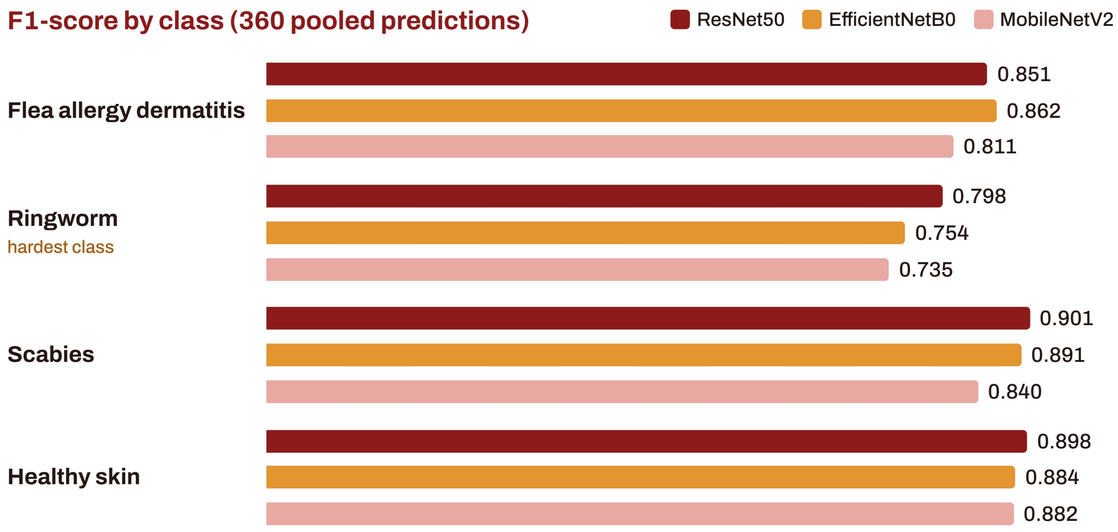
Illustrative only: one correctly classified photo per class.

RESULTS

Overall performance (5-fold cross-validation, mean ± SD)

ResNet50	EfficientNetB0	MobileNetV2
86.11%	**84.72%**	**81.67%**
± 4.12 accuracy	± 3.83 accuracy	± 5.08 accuracy
Macro-F1 **0.858** · MCC **0.820**	Macro-F1 **0.846** · MCC **0.801**	Macro-F1 **0.815** · MCC **0.761**

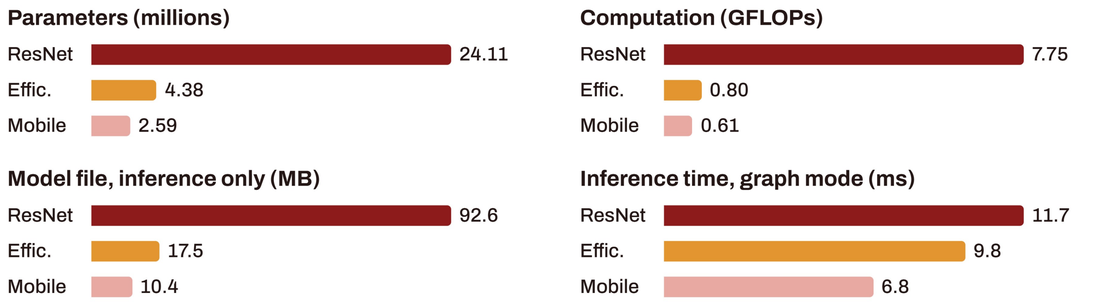
F1-score by class (360 pooled predictions)



Disease-vs-healthy screening

	ResNet50	EfficientNetB0	MobileNetV2
Diseased photos called healthy (of 270)	**7**	11	14
Healthy photos called diseased (of 90)	11	10	**8**
Screening sensitivity	0.974	0.959	0.948

Computing cost (lower is lighter)



Statistical test: paired McNemar tests found **no significant accuracy difference** between any two models after Holm correction (adjusted p = 0.113, 0.306, 0.560).

CONCLUSION & RECOMMENDATIONS

No model was best on every criterion. **ResNet50** had the highest observed accuracy and missed the fewest diseased cats, but it is by far the heaviest. **EfficientNetB0** came close with about one-fifth of the parameters and one-tenth of the FLOPs. **MobileNetV2** was the smallest and fastest but least accurate. Since no accuracy gap was significant, the clearest differences are in computing cost, so the right choice depends on the device and purpose. The models are screening aids, not diagnostic tools.

- Collect more and clearer vet-verified photos, especially of ringworm.
- Repeat with several random seeds and test on clinic photos.
- Tune each model and time it on the actual target device.

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